

Minimax Lower Bound for Multi-Expert Noise Detection: Hellinger Distance Proof

Purpose: Rewrite the minimax lower bound proof using Hellinger distance tensorization, replacing the invalid Slud-inequality approach from v1. All three issues identified in the review (`review_minimax_lower_bound.md`) are resolved.

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Fixed: 2026-06-27 — Corrected F1 bound (Sections 7.3, 7.5) and $K > 2$ Hellinger formula (Section 8.3). See `review_minimax_v2.md` for details of each bug.

Method: Hellinger distance + tensorization + Le Cam's two-point method

Key advantage over v1: Hellinger distance tensorizes exactly under product distributions ($H^2(P^{\otimes M}, Q^{\otimes M}) = 1 - (1 - H^2(P, Q))^M$), giving a clean exponential bound without requiring Slud's inequality or Gaussian approximations.

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1. Setup and Main Result

1.1 The Statistical Problem

We consider the multi-expert noise detection problem from Theorem 1. Fix a state s with clean error rate $\mu_s \in (0, 1/2)$. For a given sample x , we observe the vector of expert error indicators $(e_1, \dots, e_M)$ where $e_m = \mathbf{1}\{f_m(x) \neq y\}$, and y is the observed label.

We consider two hypotheses: - H_0 (**clean**): $y = y^*$ (the true label). Each expert makes an error with probability μ_s , independent of other experts given x . - H_1 (**noise**): $y \neq y^*$ (the label is flipped). For binary classification ($K = 2$), each expert makes an error with probability $1 - \mu_s$, independent of other experts given the noise label.

Under $K = 2$, both hypotheses yield product distributions:

$$P_0 = \text{Bernoulli}(\mu_s)^{\otimes M}, \quad P_1 = \text{Bernoulli}(1 - \mu_s)^{\otimes M}$$

The separation gap is $\Delta = \frac{1}{2} - \mu_s$, and the mean separation is 2Δ .

1.2 Main Result

Theorem 4 (Minimax Lower Bound for Multi-Expert Noise Detection). Assume (A1)-(A6). Fix a state s with clean error rate $\mu_s \in (0, 1/2)$, expert count $M \geq 1$, noise rate $\eta \in (0, 1/2)$, and class count $K \geq 2$. Let ψ be any measurable noise detector (a function of $(e_1, \dots, e_M)$).

Define the testing error:

$$R(\psi) = \max\{\mathbb{P}_0(\psi = 1), \mathbb{P}_1(\psi = 0)\}$$

where $\mathbb{P}_0$ is the distribution under “clean” and $\mathbb{P}_1$ under “noise”.

Part (a) – Testing lower bound. For $K = 2$:

$$\inf_{\psi} \sup_{P \in \mathcal{P}_{\Delta}} R(\psi) \geq \frac{1}{4} (2\sqrt{\mu_s(1 - \mu_s)})^M \geq \frac{1}{4} \exp(-2M\Delta^2) \cdot (1 - O(M\Delta^4))$$

Part (b) – F1 lower bound. Under the same conditions:

$$\inf_{\psi} \sup_{P \in \mathcal{P}_{\Delta}} [1 - \text{F1}(\psi, P)] \geq \frac{\rho^M}{8 - \rho^M}$$

where $\rho = 2\sqrt{\mu_s(1 - \mu_s)}$. For small ρ^M (equivalently, small Δ):

$$\inf_{\psi} \sup_{P \in \mathcal{P}_{\Delta}} [1 - \text{F1}(\psi, P)] \geq \frac{1}{8} (2\sqrt{\mu_s(1 - \mu_s)})^M \geq \frac{1}{8} \exp(-2M\Delta^2 + O(M\Delta^4))$$

Part (c) – Rate optimality. The SCX consistency detector of Theorem 1 achieves $1 - \text{F1} \leq \frac{1}{\eta} \exp(-2M\Delta^2)$. By Part (b), no detector can achieve exponent greater than $2M\Delta^2$ in the small-gap regime. Therefore SCX is **minimax rate-optimal** in the M -regime.

2. Hellinger Distance for One Expert

2.1 Definition and Properties

The **Hellinger distance** between two probability distributions P and Q on a discrete space is:

$$H(P, Q) = \sqrt{\frac{1}{2} \sum_x (\sqrt{P(x)} - \sqrt{Q(x)})^2}$$

Equivalently, the **Hellinger affinity** (or Bhattacharyya coefficient) is:

$$\rho(P, Q) = \sum_x \sqrt{P(x)Q(x)}$$

and:

$$H^2(P, Q) = 1 - \rho(P, Q)$$

2.2 Hellinger Distance for Bernoulli Distributions

For a single expert under $K = 2$:

$$P_{\text{clean}} = \text{Bernoulli}(\mu_s), \quad P_{\text{noise}} = \text{Bernoulli}(1 - \mu_s)$$

The Hellinger affinity is:

$$\begin{aligned} \rho &= \sum_{e \in \{0,1\}} \sqrt{P_{\text{clean}}(e) \cdot P_{\text{noise}}(e)} \\ &= \sqrt{\mu_s \cdot (1 - \mu_s)} + \sqrt{(1 - \mu_s) \cdot \mu_s} \\ &= 2\sqrt{\mu_s(1 - \mu_s)} \end{aligned}$$

The squared Hellinger distance is:

$$H^2(P_{\text{clean}}, P_{\text{noise}}) = 1 - \rho = 1 - 2\sqrt{\mu_s(1 - \mu_s)}$$

Key observation: $\rho < 1$ whenever $\mu_s \neq 1/2$, and ρ decreases as μ_s moves away from $1/2$. When $\mu_s = 1/2$ (experts at chance), $\rho = 1$ and $H = 0$ (the distributions are identical).

2.3 Example Values

μ_s	Δ	$\rho = 2\sqrt{\mu_s(1 - \mu_s)}$	H^2
0.20	0.30	$2\sqrt{0.16} = 0.800$	0.200
0.25	0.25	$2\sqrt{0.1875} = 0.866$	0.134
0.30	0.20	$2\sqrt{0.21} = 0.917$	0.083
0.40	0.10	$2\sqrt{0.24} = 0.980$	0.020

μ_s	Δ	$\rho = 2\sqrt{\mu_s(1 - \mu_s)}$	H^2
0.45	0.05	$2\sqrt{0.2475} = 0.995$	0.005

3. Tensorization Across M Experts

3.1 The Tensorization Property

A fundamental property of Hellinger distance is its behavior under product measures. For product distributions $P^{\otimes M}$ and $Q^{\otimes M}$ (where P and Q are distributions on the same space, and all M coordinates are i.i.d.):

$$1 - H^2(P^{\otimes M}, Q^{\otimes M}) = (1 - H^2(P, Q))^M$$

Equivalently, in terms of the Hellinger affinity $\rho = 1 - H^2(P, Q)$:

$$\rho_M = \rho^M, \quad H^2(P^{\otimes M}, Q^{\otimes M}) = 1 - \rho^M$$

Proof. By definition of the Hellinger affinity for product measures:

$$\begin{aligned}
 \rho(P^{\otimes M}, Q^{\otimes M}) &= \sum_{e_1, \dots, e_M} \sqrt{\prod_{m=1}^M P(e_m) \cdot \prod_{m=1}^M Q(e_m)} \\
 &= \sum_{e_1, \dots, e_M} \prod_{m=1}^M \sqrt{P(e_m)Q(e_m)} \\
 &= \prod_{m=1}^M \sum_{e \in \{0,1\}} \sqrt{P(e)Q(e)} \\
 &= (\sqrt{P(0)Q(0)} + \sqrt{P(1)Q(1)})^M \\
 &= \rho^M
 \end{aligned}$$

The squared Hellinger distance follows: $H^2(P^{\otimes M}, Q^{\otimes M}) = 1 - \rho^M$. Note that this tensorization is exact – there is no slack or inequality.

3.2 Application to the Multi-Expert Problem

For $K = 2$, the clean and noise distributions are product distributions:

$$P_0 = \text{Bernoulli}(\mu_s)^{\otimes M}, \quad P_1 = \text{Bernoulli}(1 - \mu_s)^{\otimes M}$$

The Hellinger affinity for the M -expert problem is:

$$\rho_M = (2\sqrt{\mu_s(1 - \mu_s)})^M$$

The squared Hellinger distance is:

$$H^2(P_0, P_1) = 1 - (2\sqrt{\mu_s(1 - \mu_s)})^M$$

Why this works: The conditional independence assumption (A2) ensures that given x , the expert errors are independent. Under $K = 2$, the noise distribution has only one component (the single wrong class), so it is a pure product distribution, not a mixture. This makes tensorization exact.

4. Total Variation Bound

4.1 The TV-Hellinger Inequality

Lemma 1 (TV-Hellinger bound). For any two probability distributions P and Q :

$$\text{TV}(P, Q) \leq H(P, Q)$$

where $\text{TV}(P, Q) = \frac{1}{2} \sum_x |P(x) - Q(x)|$.

Proof. Using the Cauchy-Schwarz inequality:

$$\begin{aligned} 2 \cdot \text{TV}(P, Q) &= \sum_x |P(x) - Q(x)| \\ &= \sum_x |\sqrt{P(x)} - \sqrt{Q(x)}| \cdot |\sqrt{P(x)} + \sqrt{Q(x)}| \\ &\leq \sqrt{\sum_x (\sqrt{P(x)} - \sqrt{Q(x)})^2} \cdot \sqrt{\sum_x (\sqrt{P(x)} + \sqrt{Q(x)})^2} \\ &= \sqrt{2H^2(P, Q)} \cdot \sqrt{2 + 2\rho(P, Q)} \\ &\leq \sqrt{2H^2(P, Q)} \cdot \sqrt{4} = 2\sqrt{2} \cdot H(P, Q) \end{aligned}$$

This gives $\text{TV} \leq \sqrt{2} \cdot H$, but a sharper bound is known: $\text{TV} \leq H$. The standard proof uses the inequality $|a - b| \leq \sqrt{|a^2 - b^2|}$ for $a, b \geq 0$:

$$\begin{aligned}
\text{TV}(P, Q) &= \frac{1}{2} \sum_x |P(x) - Q(x)| \\
&= \frac{1}{2} \sum_x |\sqrt{P(x)} - \sqrt{Q(x)}| \cdot |\sqrt{P(x)} + \sqrt{Q(x)}| \\
&\leq \frac{1}{2} \sqrt{\sum_x (\sqrt{P(x)} - \sqrt{Q(x)})^2 \cdot \sum_x (\sqrt{P(x)} + \sqrt{Q(x)})^2} \\
&= \frac{1}{2} \sqrt{2H^2 \cdot 2(1 + \rho)} = H \cdot \sqrt{\frac{1 + \rho}{2}} \leq H
\end{aligned}$$

where the last inequality uses $\rho \leq 1$. $\square$

4.2 Applying the Bound

Applying Lemma 1 to the M -expert problem:

$$\text{TV}(P_0, P_1) \leq H(P_0, P_1) = \sqrt{1 - (2\sqrt{\mu_s(1 - \mu_s)})^M}$$

Let $\rho = 2\sqrt{\mu_s(1 - \mu_s)}$. Then:

$$\text{TV}(P_0, P_1) \leq \sqrt{1 - \rho^M}$$

4.3 Comparison with Other Bounds

The TV-Hellinger bound is tighter than the TV- χ^2 bound ($\text{TV} \leq \sqrt{\chi^2/2}$) in the following sense:

- **Hellinger** bounds TV by $\sqrt{1 - \rho^M}$, where ρ is the per-expert Hellinger affinity
- χ^2 bounds TV by $\sqrt{((1 + \chi_{\text{bern}}^2)^M - 1)/2}$ where $\chi_{\text{bern}}^2 = (1 - 2\mu_s)^2/(\mu_s(1 - \mu_s))$

The Hellinger bound is cleaner and avoids the $1/2$ factor inside the square root.

5. Le Cam Two-Point Lower Bound

5.1 Le Cam's Lemma

Lemma 2 (Le Cam, 1973). Let P_0 and P_1 be two probability distributions on the same measurable space. For any test $\psi \in \{0, 1\}$:

$$\mathbb{P}_0(\psi = 1) + \mathbb{P}_1(\psi = 0) \geq 1 - \text{TV}(P_0, P_1)$$

Equivalently:

$$\max\{\mathbb{P}_0(\psi = 1), \mathbb{P}_1(\psi = 0)\} \geq \frac{1 - \text{TV}(P_0, P_1)}{2}$$

Proof. For any test ψ :

$$\begin{aligned}
 \mathbb{P}_0(\psi = 1) + \mathbb{P}_1(\psi = 0) &= 1 - (\mathbb{P}_0(\psi = 0) - \mathbb{P}_1(\psi = 0)) \\
 &= 1 - (\mathbb{P}_1(\psi = 0) - \mathbb{P}_0(\psi = 0)) \\
 &= 1 - \int \psi \cdot (dP_0 - dP_1) \quad (\text{depending on the test's form}) \\
 &\geq 1 - \text{TV}(P_0, P_1)
 \end{aligned}$$

The last inequality uses the definition of TV as the maximum difference in probability over measurable sets. $\square$

5.2 Applying Le Cam with the Hellinger TV Bound

Combining Lemma 2 with the TV bound from Section 4:

$$\begin{aligned}
 R(\psi) &\geq \frac{1 - \text{TV}(P_0, P_1)}{2} \\
 &\geq \frac{1 - \sqrt{1 - \rho^M}}{2}
 \end{aligned}$$

where $\rho = 2\sqrt{\mu_s(1 - \mu_s)}$.

5.3 Simplifying the Bound

Lemma 3 (Useful inequality). For $x \in [0, 1]$:

$$1 - \sqrt{1 - x} \geq \frac{x}{2}$$

Proof. Let $f(x) = 1 - \sqrt{1 - x} - x/2$. Then $f(0) = 0$, and:

$$f'(x) = \frac{1}{2\sqrt{1-x}} - \frac{1}{2} \geq 0 \quad \text{for } x \in [0, 1]$$

since $\sqrt{1-x} \leq 1$. Thus $f(x) \geq 0$. $\square$

Applying Lemma 3 with $x = \rho^M$:

$$R(\psi) \geq \frac{\rho^M}{4} = \frac{1}{4} (2\sqrt{\mu_s(1-\mu_s)})^M$$

This gives a clean, explicit lower bound on the minimax testing error.

Why this is correct: Every inequality direction has been verified: - Hellinger affinity $\rho \leq 1$ (holds for any distributions) - $\text{TV} \leq H$ (standard, direction is correct for lower bound: larger H gives larger upper bound on TV, which gives smaller lower bound on error, so this makes the bound conservative) - $1 - \sqrt{1-x} \geq x/2$ (verified, direction is correct: we're replacing the true value with a smaller value, making the lower bound weaker but still valid) - The final bound $R(\psi) \geq \rho^M/4$ is a valid (conservative) lower bound

6. Expression in Terms of the Separation Gap

6.1 The Separation Gap Δ

For $K = 2$ with clean error rate μ_s , the separation gap is:

$$\mu_s = \frac{1}{2} - \Delta, \quad \text{where } \Delta \in (0, 1/2)$$

6.2 Expressing ρ in Terms of Δ

$$\begin{aligned}
\rho &= 2\sqrt{\mu_s(1 - \mu_s)} \\
&= 2\sqrt{\left(\frac{1}{2} - \Delta\right)\left(\frac{1}{2} + \Delta\right)} \\
&= 2\sqrt{\frac{1}{4} - \Delta^2} \\
&= \sqrt{1 - 4\Delta^2}
\end{aligned}$$

6.3 The Lower Bound in Terms of Δ

$$R(\psi) \geq \frac{1}{4}(\sqrt{1 - 4\Delta^2})^M = \frac{1}{4}(1 - 4\Delta^2)^{M/2}$$

6.4 Small- Δ Approximation

For small Δ , using $\log(1 - 4\Delta^2) = -4\Delta^2 - 8\Delta^4/3 - O(\Delta^6)$:

$$\begin{aligned}
R(\psi) &\geq \frac{1}{4} \exp\left(\frac{M}{2} \log(1 - 4\Delta^2)\right) \\
&= \frac{1}{4} \exp\left(-2M\Delta^2 - \frac{4}{3}M\Delta^4 - O(M\Delta^6)\right) \\
&\geq \frac{1}{4} \exp(-2M\Delta^2) \cdot (1 - O(M\Delta^4))
\end{aligned}$$

The leading exponent is $2M\Delta^2$. This matches the Hoeffding exponent in Theorem 1's upper bound.

6.5 Alternative Expression (Exact, No Approximation)

For any Δ (not necessarily small):

$$R(\psi) \geq \frac{1}{4}(1 - 4\Delta^2)^{M/2}$$

This is the **exact** exponential form before any approximation. It is valid for all $M \geq 1$ and $\Delta \in [0, 1/2)$, with no ceiling issues or parameter restrictions.

7. F1 Lower Bound

7.1 From Testing Error to F1

The F1 score is defined at the population level:

$$\text{F1} = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}$$

where: - $\text{TP} = \eta \cdot \mathbb{P}(\psi = 1 \mid \text{noise}) = \eta \cdot (1 - \beta)$ - $\text{FP} = (1 - \eta) \cdot \mathbb{P}(\psi = 1 \mid \text{clean}) = (1 - \eta) \cdot \alpha$ - $\text{FN} = \eta \cdot \mathbb{P}(\psi = 0 \mid \text{noise}) = \eta \cdot \beta$

with $\alpha = P_0(\psi = 1)$ and $\beta = P_1(\psi = 0)$.

7.2 Upper Bound on F1 from the Testing Bound

From Section 5, we have $R(\psi) = \max(\alpha, \beta) \geq \varepsilon$ where $\varepsilon = \rho^M/4$.

To convert this to an F1 upper bound, note that F1 is decreasing in both α and β . The maximum possible F1 under the constraint $\max(\alpha, \beta) \geq \varepsilon$ is achieved at the boundary where exactly one of α or β equals ε and the other is zero:

Scenario A ($\beta = \varepsilon, \alpha = 0$):

$$\text{F1} = \frac{2\eta(1-\varepsilon)}{2\eta(1-\varepsilon) + 0 + \eta\varepsilon} = \frac{2(1-\varepsilon)}{2-\varepsilon}$$

Scenario B ($\alpha = \varepsilon, \beta = 0$):

$$\text{F1} = \frac{2\eta}{2\eta + (1-\eta)\varepsilon + 0} = \frac{2\eta}{2\eta + (1-\eta)\varepsilon}$$

For $\eta \leq 1/2$, Scenario B yields a smaller (stricter) upper bound:

$$\frac{2\eta}{2\eta + (1-\eta)\varepsilon} \leq \frac{2(1-\varepsilon)}{2-\varepsilon} \quad \text{for all } \varepsilon \in [0, 1]$$

(The inequality holds because $2\eta/(2\eta + (1-\eta)\varepsilon) \leq 2(1-\varepsilon)/(2-\varepsilon)$ is equivalent to $\varepsilon(1-2\eta) + \eta\varepsilon^2 \geq 0$, which is true for $\eta \leq 1/2$.)

Thus:

$$\text{F1} \leq \frac{2\eta}{2\eta + (1-\eta)\varepsilon}$$

and:

$$1 - \text{F1} \geq \frac{(1-\eta)\varepsilon}{2\eta + (1-\eta)\varepsilon}$$

7.3 The Universal F1 Lower Bound

From the case analysis in Section 7.5, the universal bound is:

$$1 - \text{F1} \geq \frac{\varepsilon}{2 - \varepsilon}$$

Using $\varepsilon \geq \rho^M/4$ and noting that the RHS is increasing in ε :

$$1 - \text{F1} \geq \frac{\rho^M/4}{2 - \rho^M/4} = \frac{\rho^M}{8 - \rho^M}$$

For small ρ^M (which occurs for any fixed $\mu_s < 1/2$ when M is sufficiently large, or for fixed M when μ_s is sufficiently far from $1/2$):

$$1 - \text{F1} \geq \frac{\rho^M}{8} = \frac{1}{8} (2\sqrt{\mu_s(1 - \mu_s)})^M$$

Note: This bound is independent of η (no η in the leading constant $1/8$), which is a stronger statement than the earlier incorrect derivation that introduced $1/(16\eta)$. The exponent $-\log \rho$ is unaffected by this correction.

7.4 Expression in Terms of Δ

$$1 - \text{F1} \geq \frac{1}{8} (1 - 4\Delta^2)^{M/2}$$

For small Δ :

$$1 - \text{F1} \geq \frac{1}{8} \exp(-2M\Delta^2 - 4M\Delta^4 - O(M\Delta^6))$$

7.5 Verification of the F1 Bound Direction

Claim: $1 - \text{F1} \geq \frac{\varepsilon}{2 - \varepsilon}$ when $\max(\alpha, \beta) \geq \varepsilon$.

Proof. If $\max(\alpha, \beta) \geq \varepsilon$, then either $\beta \geq \varepsilon$ or $\alpha \geq \varepsilon$ (or both).

- **Case 1** ($\beta \geq \varepsilon$): Then $\text{FN} = \eta\beta \geq \eta\varepsilon$. Using $\text{F1} \leq \frac{2\text{TP}}{2\text{TP} + \text{FN}}$ (since $\text{FP} \geq 0$):

$$\text{F1} \leq \frac{2\eta(1 - \beta)}{2\eta(1 - \beta) + \eta\beta} \leq \frac{2\eta(1 - \varepsilon)}{2\eta(1 - \varepsilon) + \eta\varepsilon} = \frac{2(1 - \varepsilon)}{2 - \varepsilon}$$

Thus $1 - \text{F1} \geq \frac{\varepsilon}{2 - \varepsilon}$.

- **Case 2** ($\alpha \geq \varepsilon$): Then $\text{FP} = (1 - \eta)\alpha \geq (1 - \eta)\varepsilon$. Using $\text{F1} \leq \frac{2\text{TP}}{2\text{TP} + \text{FP}}$ (since $\text{FN} \geq 0$):

$$\text{F1} \leq \frac{2\eta}{2\eta + (1 - \eta)\alpha} \leq \frac{2\eta}{2\eta + (1 - \eta)\varepsilon}$$

Thus $1 - \text{F1} \geq \frac{(1 - \eta)\varepsilon}{2\eta + (1 - \eta)\varepsilon}$.

Since $\max(\alpha, \beta) \geq \varepsilon$, at least one case must hold. The universal bound is the minimum of the two case-specific bounds:

$$1 - \text{F1} \geq \min\left(\frac{\varepsilon}{2 - \varepsilon}, \frac{(1 - \eta)\varepsilon}{2\eta + (1 - \eta)\varepsilon}\right)$$

For $\eta \leq 1/2$ and small ε (the regime of interest), compare the approximate magnitudes: $\frac{\varepsilon}{2 - \varepsilon} \approx \frac{\varepsilon}{2}$ and $\frac{(1 - \eta)\varepsilon}{2\eta + (1 - \eta)\varepsilon} \approx \frac{(1 - \eta)\varepsilon}{2\eta}$. The ratio is $\frac{\eta}{1 - \eta} \leq 1$, so Case~1 gives the smaller (tighter) bound. Hence:

$$1 - \text{F1} \geq \frac{\varepsilon}{2 - \varepsilon}$$

This bound is universal: it holds regardless of whether the test falls into Case~1 or Case~2, because it is the smaller of the two case-specific bounds. No cross-multiplication or parameter restriction is needed. $\square$

8. Extension to $K > 2$

8.1 The Mixture Structure

For $K > 2$, the noise distribution becomes a mixture over $(K - 1)$ wrong classes:

$$P_1 = \frac{1}{K-1} \sum_{c \neq y^*} \text{Bernoulli}\left(1 - \frac{\mu_s}{K-1}\right)^{\otimes M}$$

where under assumption A6 with $C_{\text{bal}} = 1$, all mixture components have the same Bernoulli parameter $p_c = 1 - \mu_s/(K - 1)$.

8.2 Reduction to Binary Case

Lemma 4 (K=2 is the hardest case under A6 with C_bal=1). When all mixture components are identical, the noise distribution P_1 is itself a product distribution:

$$P_1 = \text{Bernoulli}\left(1 - \frac{\mu_s}{K-1}\right)^{\otimes M}$$

Proof. If all $K - 1$ mixture components have the same parameter, the mixture is trivial:

$$P_1 = \frac{1}{K-1} \cdot (K-1) \cdot \text{Bernoulli}(1-\mu_s/(K-1))^{\otimes M} = \text{Bernoulli}(1-\mu_s/(K-1))^{\otimes M}$$

Thus P_1 is a product distribution, and the Hellinger approach applies directly.

□

> **FIXED (2026-06-28, DEFECT-16):** Lemma 4’s claim “ $K=2$ is the hardest case” is proven **only under** $C_{\text{bal}} = 1$ (perfectly balanced error distribution across wrong classes). When $C_{\text{bal}} = 1$, all $K-1$ mixture components have identical Bernoulli parameters, the mixture collapses to a single product distribution, and $K=2$ gives the worst-case (largest) per-expert Hellinger affinity $\rho_2 = 2\sqrt{\mu_s(1-\mu_s)}$.

For $C_{\text{bal}} > 1$, the mixture components are non-identical. The worst-case component (closest to the clean distribution) has error probability $p_1 = 1 - C_{\text{bal}} \cdot \mu_s/(K-1)$, which is **smaller** than the $C_{\text{bal}} = 1$ case. Smaller p_1 means larger separation from $p_0 = \mu_s$, making the testing problem **easier**, not harder. Therefore:

- **The minimax rate** $\kappa_{\min} = \min_s \kappa_s$ (**exponent in M**) **is dominated by the** $C_{\text{bal}} = 1$ **case**, since larger C_{bal} produces larger separation gaps.
- **The exact minimax constant** $C_{\min}$ **for** $C_{\text{bal}} > 1$ involves the worst-case mixture component via the convexity bound of Section 9.2, and may differ from the $C_{\text{bal}} = 1$ constant by a factor depending on $\lambda_1^*(C_{\text{bal}})$.
- **The general** $C_{\text{bal}} > 1$ **case for the exact constant remains partially open**, though numerical evidence (cases 1-5 in the verification table) suggests the constant matching between SCX achievability and the minimax lower bound holds for practical values $C_{\text{bal}} \in [1, 3]$.

Practical implication: The $K=2 + C_{\text{bal}} = 1$ case provides the **most stringent test** of the theory. Any noise detection guarantee that holds for this hardest case automatically holds for easier cases ($C_{\text{bal}} > 1$, $K > 2$ with balanced errors). The SCX minimax optimality claim in Theorem 4' should explicitly note the $C_{\text{bal}} = 1$ restriction for the exact constant, while the rate optimality (κ) is unaffected.

8.3 Comparing the Separation

For $K > 2$ with $C_{\text{bal}} = 1$, the clean and noise Bernoulli parameters are:

$$p_0 = \mu_s, \quad p_1 = 1 - \frac{\mu_s}{K-1}$$

The mean separation is:

$$p_1 - p_0 = 1 - \frac{\mu_s}{K-1} - \mu_s = 1 - \mu_s \cdot \frac{K}{K-1}$$

This is increasing in K (for fixed μ_s). The per-expert Hellinger affinity for $\text{Bernoulli}(p_0)$ vs $\text{Bernoulli}(p_1)$ is:

$$\rho_K = \sqrt{\mu_s \left(1 - \frac{\mu_s}{K-1}\right)} + \sqrt{(1 - \mu_s) \left(\frac{\mu_s}{K-1}\right)}$$

This is the correct general formula; the expression $2\sqrt{\mu_s(1 - \mu_s)}$ is only valid for $K = 2$ where $p_1 = 1 - p_0$. One can verify $\rho_K \leq 1$ via Cauchy-Schwarz: $\sqrt{pq} + \sqrt{(1-p)(1-q)} \leq 1$, with equality iff $p = q$.

Since ρ_K decreases with K (larger K means larger separation), and the testing lower bound is $\rho_K^M/4$, we have:

- For $K = 2$: $\rho_2 = 2\sqrt{\mu_s(1 - \mu_s)}$
- For $K > 2$: $\rho_K = \sqrt{\mu_s(1 - \mu_s/(K - 1))} + \sqrt{(1 - \mu_s)\mu_s/(K - 1)} < \rho_2$

Therefore $\rho_K^M < \rho_2^M$, meaning the bound for $K > 2$ is larger (harder to distinguish). Since we want a lower bound on the minimax risk, a larger bound from $K > 2$ would give a weaker guarantee. We use the $K = 2$ bound as the worst case:

$$R(\psi) \geq \frac{1}{4}\rho_2^M = \frac{1}{4}(2\sqrt{\mu_s(1 - \mu_s)})^M$$

This holds for all $K \geq 2$ under $C_{\text{bal}} = 1$.

Note on the mixture-product issue: The earlier proof (v1) attempted to bound this via χ^2 divergence and confused the direction of the bound. The Hellinger approach avoids this issue entirely because under $C_{\text{bal}} = 1$, P_1 is exactly a product distribution, and the tensorization is exact.

9. Extension to $C_{\text{bal}} > 1$

9.1 Non-Uniform Error Distribution

When $C_{\text{bal}} > 1$, the mixture components may have different Bernoulli parameters. The worst-case component (giving the largest per-expert Hellinger affinity, i.e., the most similarity to the clean distribution) has error probability:

$$p_{\max} = \max_{c \neq y^*} \left(1 - \frac{\mu_s}{K - 1} \cdot C_{\text{bal}} \right)$$

9.2 Hellinger Bound for Mixtures

For the mixture $P_1 = \frac{1}{L} \sum_{\ell=1}^L Q_\ell$ where Q_ℓ are product distributions, we can bound the Hellinger distance using convexity.

Lemma 5 (Hellinger convexity for mixtures). Let P_0 be a distribution and $P_1 = \frac{1}{L} \sum_{\ell=1}^L Q_\ell$ be a mixture. Then:

$$H^2(P_0, P_1) \geq \frac{1}{L} \sum_{\ell=1}^L H^2(P_0, Q_\ell) - \frac{1}{L} \sum_{\ell < \ell'} \rho(Q_\ell, Q_{\ell'})$$

Proof sketch. The squared Hellinger distance is not convex in its second argument, so the standard convexity bound for TV does not apply. Instead, we use the affinity:

$$\rho(P_0, P_1) = \sum_x \sqrt{P_0(x) \cdot \frac{1}{L} \sum_{\ell} Q_\ell(x)} \geq \frac{1}{L} \sum_{\ell} \sum_x \sqrt{P_0(x) Q_\ell(x)} = \frac{1}{L} \sum_{\ell} \rho(P_0, Q_\ell)$$

by the concavity of $\sqrt{\cdot}$ and Jensen's inequality. Therefore:

$$H^2(P_0, P_1) = 1 - \rho(P_0, P_1) \leq 1 - \frac{1}{L} \sum_{\ell} \rho(P_0, Q_\ell) = \frac{1}{L} \sum_{\ell} H^2(P_0, Q_\ell)$$

Wait – this gives an **upper bound** on $H^2(P_0, P_1)$, which when combined with $\text{TV} \leq H$ would give an upper bound on TV, which is what we need for Le Cam. But let me double-check the direction.

Actually, the concavity of $\sqrt{\cdot}$ gives:

$$\rho(P_0, P_1) = \sum_x \sqrt{P_0(x) \cdot \frac{1}{L} \sum_{\ell} Q_{\ell}(x)} = \sum_x \sqrt{P_0(x)} \cdot \sqrt{\frac{1}{L} \sum_{\ell} Q_{\ell}(x)}$$

By the concavity of $\sqrt{\cdot}$: $\sqrt{\frac{1}{L} \sum_{\ell} Q_{\ell}(x)} \geq \frac{1}{L} \sum_{\ell} \sqrt{Q_{\ell}(x)}$.

Therefore:

$$\rho(P_0, P_1) \geq \sum_x \sqrt{P_0(x)} \cdot \frac{1}{L} \sum_{\ell} \sqrt{Q_{\ell}(x)} = \frac{1}{L} \sum_{\ell} \sum_x \sqrt{P_0(x) Q_{\ell}(x)} = \frac{1}{L} \sum_{\ell} \rho(P_0, Q_{\ell})$$

Thus $\rho(P_0, P_1) \geq \frac{1}{L} \sum_{\ell} \rho(P_0, Q_{\ell})$, which means:

$$H^2(P_0, P_1) = 1 - \rho(P_0, P_1) \leq 1 - \frac{1}{L} \sum_{\ell} \rho(P_0, Q_{\ell}) = \frac{1}{L} \sum_{\ell} H^2(P_0, Q_{\ell})$$

This is an **upper bound** on H^2 . Since $\text{TV} \leq H$, this gives:

$$\text{TV}(P_0, P_1) \leq H(P_0, P_1) \leq \sqrt{\frac{1}{L} \sum_{\ell} H^2(P_0, Q_{\ell})}$$

For the lower bound (via Le Cam), a smaller TV gives a stronger lower bound, so this direction is correct. However, the bound might be loose if the mixture components are very different from each other.

9.3 Practical Bound for $C_{\text{bal}} > 1$

In practice, we can use a simpler bound. Let $H_{\max}^2 = \max_{\ell} H^2(P_0, Q_{\ell})$. Then:

$$\text{TV}(P_0, P_1) \leq \sqrt{\frac{1}{L} \sum_{\ell} H^2(P_0, Q_{\ell})} \leq \sqrt{H_{\max}^2} = \max_{\ell} H(P_0, Q_{\ell})$$

Therefore the worst (largest) Hellinger distance among the mixture components gives the tightest upper bound on TV. The testing lower bound is then:

$$R(\psi) \geq \frac{1 - \max_{\ell} H(P_0, Q_{\ell})}{2}$$

For $C_{\text{bal}} > 1$, the component with the smallest $1 - C_{\text{bal}} \cdot \mu_s / (K - 1)$ (most similar to clean) gives the smallest Hellinger distance, and hence the weakest lower bound. The worst-case over ℓ for the lower bound is:

$$\min_{\ell} H^2(P_0, Q_{\ell}) = 1 - 2\sqrt{\mu_s \cdot \min_{\ell}(1 - \mu_{c_{\ell}})}$$

where $\mu_{c_{\ell}}$ is the probability that an expert predicts class c_{ℓ} under noise.

Message: For $C_{\text{bal}} > 1$, the exponent $2M\Delta^2$ is preserved, but the constant factor degrades. The rate optimality (exponent in M) is unaffected.

10. Rate Optimality

10.1 Upper Bound (Theorem 1)

From Theorem 1 (Hoeffding form):

$$1 - \text{F1} \leq \frac{1}{\eta} \sum_s \rho_s \cdot \exp(-2M\Delta_s^2)$$

For the hardest state $s^* = \arg \min_s \Delta_s$, this gives:

$$1 - \text{F1} \leq \frac{1}{\eta} \exp(-2M\Delta_{s^*}^2)$$

10.2 Lower Bound (Theorem 4, this document)

From Section 7:

$$1 - \text{F1} \geq \frac{1}{8} \exp(-2M\Delta^2 + O(M\Delta^4))$$

10.3 Optimality Statement

Quantity	Upper bound (Thm 1)	Lower bound (Thm 4)	Ratio
Exponent	$2M\Delta^2$	$2M\Delta^2$	1 (exact match)
Leading constant	$1/\eta$	$1/8$	$8/\eta$ (at least 16)
Regime	All Δ	Small Δ	Asymptotic in Δ

The exponent $2M\Delta^2$ is **minimax optimal**: no detector can achieve a better exponential rate. The constant factor gap is $8/\eta$ (at least 16 for $\eta \leq 1/2$), which is typical for minimax results and can be tightened with more refined arguments.

Corollary (Rate optimality). The SCX consistency detector achieves the minimax-optimal rate $\exp(-2M\Delta^2)$ in the small-gap regime. This is the

information-theoretic limit for multi-expert noise detection under assumptions (A1)-(A6).

11. Connection to Theorem 1: Chernoff Exactness

11.1 The Chernoff Exponent

A remarkable property of the Hellinger proof is that it naturally yields the Chernoff exponent, not just the Hoeffding exponent.

For $K = 2$, the exact large-deviations rate for the optimal Bayes test is given by the Cramer-Chernoff theorem. The optimal threshold is $\theta^* = 1/2$, and:

$$\text{KL}\left(\frac{1}{2} \parallel \mu_s\right) = \frac{1}{2} \log \frac{1}{2\mu_s} + \frac{1}{2} \log \frac{1}{2(1-\mu_s)} = -\log(2\sqrt{\mu_s(1-\mu_s)})$$

11.2 The Hellinger-Chernoff Correspondence

The per-expert Hellinger exponent is:

$$-\log \rho = -\log(2\sqrt{\mu_s(1-\mu_s)}) = \text{KL}\left(\frac{1}{2} \parallel \mu_s\right)$$

These are **exactly equal**. This means:

- The Hellinger tensorization gives the **exact** Chernoff exponent per expert
- The lower bound exponent $-\log \rho$ matches the upper bound's Chernoff exponent exactly

11.3 Taylor Expansion Comparison

For small Δ :

$$\begin{aligned}
 -\log \rho &= -\log(2\sqrt{(1/2 - \Delta)(1/2 + \Delta)}) \\
 &= -\log \sqrt{1 - 4\Delta^2} \\
 &= -\frac{1}{2} \log(1 - 4\Delta^2) \\
 &= 2\Delta^2 + \frac{4}{3}\Delta^4 + O(\Delta^6)
 \end{aligned}$$

The Hoeffding exponent $2\Delta^2$ is the leading term. The Hellinger lower bound recovers the full Chernoff rate, which is strictly larger than $2\Delta^2$ for any $\Delta > 0$.

11.4 Matching Table

Object	Value	Exponent
Per-expert Hellinger affinity ρ	$2\sqrt{\mu_s(1 - \mu_s)}$	$-\log \rho$
KL divergence $\text{KL}(1/2 \ \mu_s)$	$-\log(2\sqrt{\mu_s(1 - \mu_s)})$	$-\log \rho$
Theorem 1 Chernoff exponent	$M \cdot \text{KL}(\theta^* \ \mu_s)$	$M \cdot (-\log \rho)$
Theorem 4 Hellinger exponent	$-M \log \rho$	$M \cdot (-\log \rho)$
Hoeffding approximation	$2M\Delta^2$	$M \cdot (2\Delta^2)$

The **exact matching** between the Hellinger exponent and the Chernoff exponent demonstrates that the bound is tight at the exponent level – not just up to constants, but exactly.

12. Numerical Verification

12.1 Testing Error Lower Bound

The lower bound $R(\psi) \geq \rho^M/4$ for various parameter values:

μ_s	Δ	ρ	M	Lower bound $\rho^M/4$
0.20	0.30	0.800	10	$0.25 \cdot 0.8^{10} \approx 0.027$
0.20	0.30	0.800	20	$0.25 \cdot 0.8^{20} \approx 0.003$
0.25	0.25	0.866	10	$0.25 \cdot 0.866^{10} \approx 0.058$
0.25	0.25	0.866	20	$0.25 \cdot 0.866^{20} \approx 0.013$
0.30	0.20	0.917	10	$0.25 \cdot 0.917^{10} \approx 0.105$
0.30	0.20	0.917	20	$0.25 \cdot 0.917^{20} \approx 0.044$
0.40	0.10	0.980	10	$0.25 \cdot 0.98^{10} \approx 0.204$
0.40	0.10	0.980	50	$0.25 \cdot 0.98^{50} \approx 0.091$
0.49	0.01	0.9998	100	$0.25 \cdot 0.9998^{100} \approx 0.245$

All bounds are non-trivial (below 0.25) when ρ^M is not too close to 1, which occurs when M is large enough or μ_s is far enough from $1/2$.

12.2 F1 Lower Bound

The F1 lower bound $1 - \text{F1} \geq \rho^M/8$:

μ_s	Δ	M	$1 - \text{F1} \geq$	Implied F1 $\leq$
0.20	0.30	10	$0.8^{10}/8 \approx 0.013$	0.987
0.20	0.30	20	$0.8^{20}/8 \approx 0.001$	0.999
0.25	0.25	10	$0.866^{10}/8 \approx 0.030$	0.970
0.25	0.25	20	$0.866^{20}/8 \approx 0.007$	0.993
0.30	0.20	20	$0.917^{20}/8 \approx 0.022$	0.978
0.40	0.10	100	$0.98^{100}/8 \approx 0.017$	0.983

Note: These F1 upper bounds are loose (close to 1) because the bound only requires that $\max(\alpha, \beta) \geq \varepsilon$, not that both are large. A more refined construction could tighten this.

12.3 Verification of Key Inequalities

We verify the key inequality $1 - \sqrt{1 - x} \geq x/2$ for the relevant range:

$x = \rho^M$	ρ (typ)	$\sqrt{1 - x}$	$1 - \sqrt{1 - x}$	$x/2$
0.01	—	0.995	0.005	0.005
0.10	—	0.949	0.051	0.050
0.25	—	0.866	0.134	0.125
0.50	—	0.707	0.293	0.250
0.75	—	0.500	0.500	0.375

The inequality holds with slack increasing in x .

12.4 Comparison with Exact Optimal Test

For $K = 2$, the optimal Bayes test for equal priors is:

$$\psi^*(e_1, \dots, e_M) = \mathbf{1}\{S > M/2\}$$

where $S = \sum_{m=1}^M e_m$. The exact error of this test is:

$$R(\psi^*) = \mathbb{P}(\text{Bin}(M, \mu_s) \geq M/2)$$

For $\mu_s = 0.2$, $M = 10$: - Exact error: $\mathbb{P}(\text{Bin}(10, 0.2) \geq 5) \approx 0.033$ - Hellinger bound: $0.25 \cdot 0.8^{10} \approx 0.027$ (close to exact) - Old Slud bound (v1): claimed $0.5 \cdot \exp(-20 \cdot 0.09) = 0.5 \cdot e^{-1.8} \approx 0.083$ (incorrect – exceeds actual error)

The Hellinger bound (0.027) is slightly below the actual error (0.033), confirming it is a valid lower bound. The old Slud bound (0.083) exceeds the actual error, confirming it was invalid.

13. Resolution of Review Issues

13.1 Issue 1: Lemma 7 (Slud’s inequality) was false

Original problem: The v1 proof claimed $\mathbb{P}(\text{Bin}(M, \mu) \geq M/2) \geq \frac{1}{2} \exp(-2M\Delta^2)$, which fails numerically (e.g., $\mu = 0.4$, $M = 20$ gives LHS ≈ 0.244 , RHS ≈ 0.335).

Resolution: The Hellinger approach entirely avoids Slud’s inequality. Instead: 1. Compute the exact Hellinger affinity $\rho = 2\sqrt{\mu_s(1 - \mu_s)}$ for one

expert 2. Tensorize exactly: $\rho_M = \rho^M$ 3. Bound $\text{TV} \leq H = \sqrt{1 - \rho^M}$ 4. Apply Le Cam: $R(\psi) \geq (1 - \sqrt{1 - \rho^M})/2 \geq \rho^M/4$

All inequalities are in the correct direction, and no Binomial tail bound is required. The Hellinger approach works for any M , any μ_s , without ceiling issues or parameter restrictions.

13.2 Issue 2: $K=2$ to $K>2$ reduction used bound direction incorrectly

Original problem: The v1 proof argued that larger χ^2 for $K > 2$ implies the $K = 2$ case is hardest. This used the bound $\text{TV} \leq \sqrt{((1 + \chi^2)^M - 1)/2}$ in the wrong direction (a larger χ^2 makes the bound looser, not tighter).

Resolution: Under $C_{\text{bal}} = 1$, all mixture components are identical, so P_1 is a pure product distribution (not a true mixture). The Hellinger approach applies directly. The per-expert parameters are $p_0 = \mu_s$ and $p_1 = 1 - \mu_s/(K - 1)$, and the separation $p_1 - p_0 = 1 - \mu_s K/(K - 1)$ increases with K , making $K = 2$ the hardest case. No bounding of the mixture distribution is needed.

13.3 Issue 3: F1 derivation was algebraically unsound

Original problem: The v1 proof's Lemma 8 attempted to show $1 - \text{F1} \geq 1 - \text{TPR}$ without verifying the validity condition $\text{FP} \geq \text{FN}$.

Resolution: The new F1 derivation (Section 7) directly bounds F1 from the testing error $\max(\alpha, \beta) \geq \varepsilon$ using the monotonicity of F1 in α and β . The derivation: 1. Considers two cases ($\alpha \geq \varepsilon$ or $\beta \geq \varepsilon$) 2. Bounds F1 in each

case using elementary inequalities 3. Takes the minimum (worst case) to get the final bound 4. Verifies all inequality directions explicitly

The bound $1 - \text{F1} \geq \rho^M/8$ is valid for all $\eta \leq 1/2$ without additional conditions.

13.4 Summary of Improvements

Aspect	v1 (Slud)	v2 (Hellinger)
Core inequality	Slud's inequality (false)	Hellinger tensorization (exact)
TV bound	Via χ^2 tensorization	Via $\text{TV} \leq H$
$K > 2$	Required mixture	Product reduction ($C_{\text{bal}} = 1$)
extension	bound (wrong direction)	
F1 derivation	Algebraic errors	Case analysis with verified inequalities
Exponent	$2M\Delta^2$ (claimed, unproven)	$2M\Delta^2$ (proved, with Chernoff exactness)
Ceiling issue	Unaddressed for odd M	Nonexistent (works for all M)
Parameter restrictions	$\mu_s \leq 1/2, k/M \leq 1/2$	None ($\mu_s \in (0, 1)$ works)
Numerical validity	Fails counterexamples	Verified numerically

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Appendix A: Glossary of Key Quantities

Symbol	Definition	Meaning
μ_s	$\mathbb{P}(f_m(x) \neq y^* \mid x \in s)$	Clean-data error rate in state s
Δ	$1/2 - \mu_s$ (for $K = 2$)	Separation gap
ρ	$2\sqrt{\mu_s(1 - \mu_s)}$	Per-expert Hellinger affinity
ρ^M	$(2\sqrt{\mu_s(1 - \mu_s)})^M$	M -expert Hellinger affinity
H^2	$1 - \rho$	Squared Hellinger distance
H	$\sqrt{1 - \rho^M}$	Hellinger distance for M experts
ε	$\rho^M/4$	Error lower bound from Le Cam

Symbol	Definition	Meaning
α	$P_0(\psi = 1)$	False positive rate
β	$P_1(\psi = 0)$	False negative rate
η	$\mathbb{P}(\text{noise})$	Global noise rate

Appendix B: Direct Calculation of Hellinger for Products

For completeness, we verify the tensorization property explicitly for the M -expert case.

Let $e = (e_1, \dots, e_M)$ with $e_m \in \{0, 1\}$. Under P_0 (clean):

$$P_0(e) = \prod_{m=1}^M \mu_s^{e_m} (1 - \mu_s)^{1-e_m}$$

Under P_1 (noise, $K = 2$):

$$P_1(e) = \prod_{m=1}^M (1 - \mu_s)^{e_m} \mu_s^{1-e_m}$$

The Hellinger affinity is:

$$\begin{aligned}
\rho(P_0, P_1) &= \sum_{e \in \{0,1\}^M} \sqrt{P_0(e)P_1(e)} \\
&= \sum_e \sqrt{\prod_{m=1}^M \mu_s^{e_m} (1 - \mu_s)^{1-e_m} \cdot \prod_{m=1}^M (1 - \mu_s)^{e_m} \mu_s^{1-e_m}} \\
&= \sum_e \prod_{m=1}^M \sqrt{\mu_s^{e_m} (1 - \mu_s)^{1-e_m} \cdot (1 - \mu_s)^{e_m} \mu_s^{1-e_m}} \\
&= \sum_e \prod_{m=1}^M (\mu_s (1 - \mu_s))^{1/2} \\
&= \sum_e (\mu_s (1 - \mu_s))^{M/2} \\
&= 2^M \cdot (\mu_s (1 - \mu_s))^{M/2} \\
&= (2\sqrt{\mu_s (1 - \mu_s)})^M
\end{aligned}$$

This confirms the tensorization property. The key step is that $\sqrt{P_0(e)P_1(e)}$ factorizes into a product across experts because:

$$\sqrt{\mu_s^{e_m} (1 - \mu_s)^{1-e_m} \cdot (1 - \mu_s)^{e_m} \mu_s^{1-e_m}} = \sqrt{\mu_s (1 - \mu_s)}$$

independently of e_m . This is the crucial simplification for the $K = 2$ case: the per-expert contribution does not depend on the expert's prediction, making the sum over 2^M error patterns collapse to a product over M independent factors.

Appendix C: Why the Hellinger Approach Works and Slud's Fails

C.1 The Fundamental Issue with Slud

Slud's inequality gives a lower bound on Binomial tail probabilities in terms of the Gaussian CDF:

$$\mathbb{P}(\text{Bin}(M, p) \geq k) \geq \frac{1}{2} \Phi \left(-\frac{k - Mp}{\sqrt{Mp(1-p)}} \right)$$

To convert this to an exponential bound, one would need to bound $\Phi(-t)$ from below by $\frac{1}{2}e^{-2Mt^2}$ (in the notation of the v1 proof). The Gaussian lower tail is:

$$\Phi(-t) \geq \frac{1}{\sqrt{2\pi}} \frac{t}{1+t^2} e^{-t^2/2}$$

The exponent is $t^2/2 = M(k/M - p)^2/(2p(1-p))$. For $p = \mu_s$, $k = M/2$:

$$\frac{t^2}{2} = \frac{M(1/2 - \mu_s)^2}{2\mu_s(1 - \mu_s)} = \frac{M\Delta^2}{2\mu_s(1 - \mu_s)}$$

Since $\mu_s(1 - \mu_s) \leq 1/4$:

$$\frac{t^2}{2} \geq \frac{M\Delta^2}{2 \cdot (1/4)} = 2M\Delta^2$$

Thus the Gaussian exponent is **at least** $2M\Delta^2$, meaning the Gaussian tail decays **at least as fast** as $\exp(-2M\Delta^2)$. A lower bound on the Binomial probability using this Gaussian tail would give an exponent of $2M\Delta^2$ or

larger – but a larger exponent means a **smaller** lower bound. The v1 proof attempted to replace this with a **larger** lower bound (using the Hoeffding exponent $2M\Delta^2$), which is the wrong direction.

Numerically: for $\mu = 0.4$, $M = 20$, the Gaussian tail $\Phi(-(10 - 8)/\sqrt{20 \cdot 0.4 \cdot 0.6}) = \Phi(-2/\sqrt{4.8}) = \Phi(-0.913) \approx 0.181$. Slud gives $\mathbb{P}(\text{Bin}(20, 0.4) \geq 10) \geq 0.181/2 = 0.090$, not $0.5 \cdot e^{-0.4} \approx 0.335$ as claimed in v1.

C.2 Why Hellinger Avoids This

The Hellinger approach directly computes the divergence between P_0 and P_1 without going through Binomial tail bounds:

1. **Exact computation:** $\rho = 2\sqrt{\mu_s(1 - \mu_s)}$ is computed directly from the Bernoulli densities, not via a tail probability bound
2. **Exact tensorization:** ρ^M is computed by multiplying per-expert affinities, not by bounding Binomial tail sums
3. **Conservative bound:** $1 - \sqrt{1 - x} \geq x/2$ relaxes the exact Le Cam bound to a simpler form, and the direction is verified to be correct (the bound is a valid lower bound, not an overestimate)

No tail bounds, no Gaussian approximations, no ceiling issues. The Hellinger approach is “robust by design” because it works at the level of densities rather than tail events.

C.3 The Ceiling Issue (Odd M)

The v1 proof applied Slud with $k = \lceil M/2 \rceil$, but Slud requires $k/M \leq 1/2$. For odd M , $\lceil M/2 \rceil/M > 1/2$, violating the premise. The Hellinger approach

has no such constraint: it works for any $M \geq 1$ because it never references thresholds or tail events.

End of proof document.